



PES
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COMPUTER NETWORKS

TEAM NETWORKS

Department of Computer Science and Engineering

Unit – 4 Network Layer and Internet Protocol

4.1 Overview of Network Layer

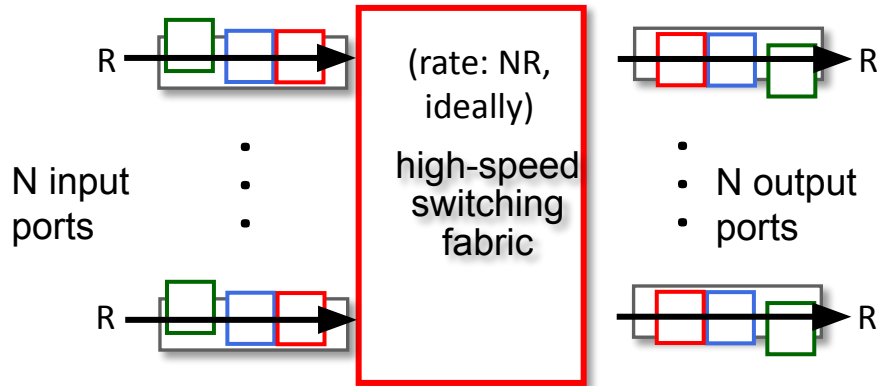
4.2 What's Inside a Router?

4.3 Switching

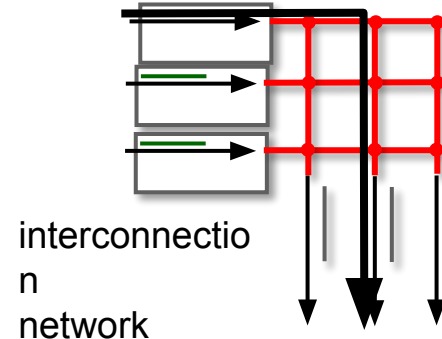
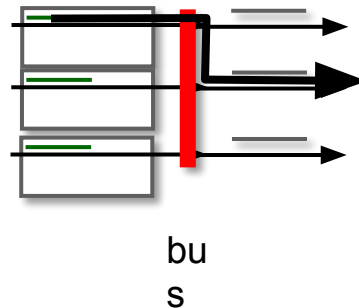
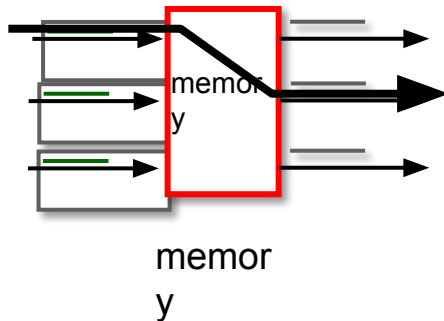
4.4 The Internet Protocol (IP)

- Datagram format
- Fragmentation
- IPv4 addressing
- NAT

- **switching rate:** rate at which packets can be transfer from inputs to outputs
 - often measured as multiple of input/output line rate
 - N inputs: switching rate N times line rate desirable

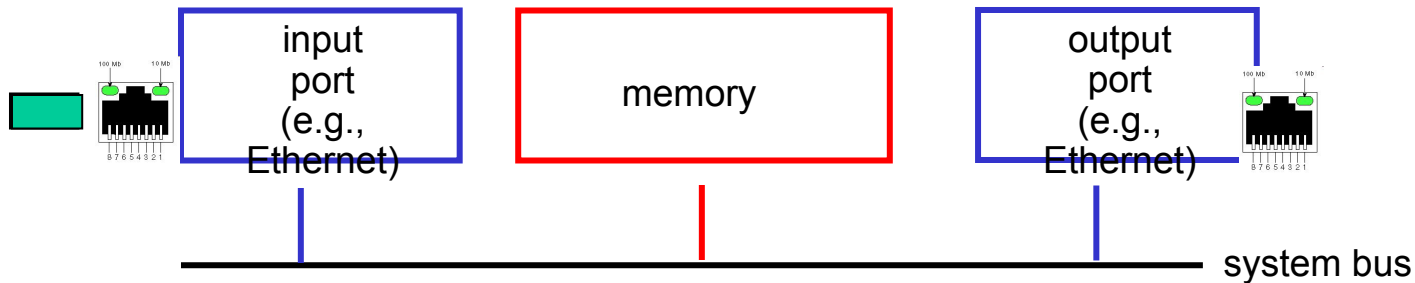


- transfer packet from input link to appropriate output link
- **switching rate:** rate at which packets can be transfer from inputs to outputs
 - often measured as multiple of input/output line rate
 - N inputs: switching rate N times line rate desirable
- three major types of switching fabrics:



first generation routers:

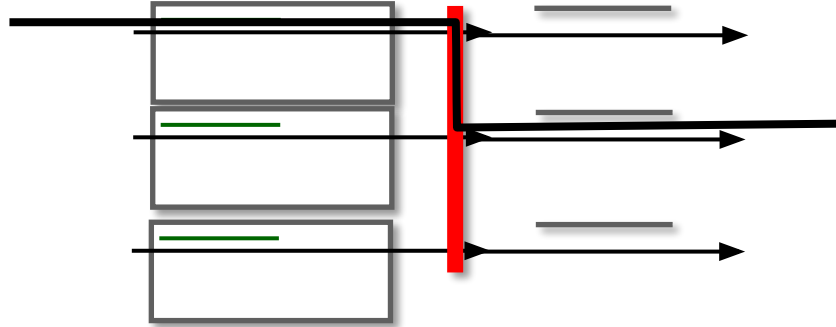
- traditional computers with switching under direct control of CPU
- packet copied to system's memory
- speed limited by memory bandwidth (2 bus crossings per datagram)



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Switching via a Bus

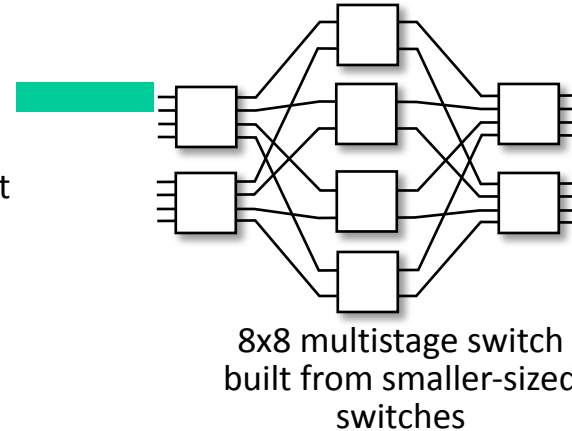
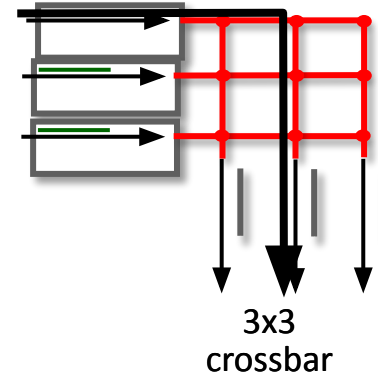
- datagram from input port memory to output port memory via a shared bus
- *bus contention*: switching speed limited by bus bandwidth
- 32 Gbps bus, Cisco 5600: sufficient speed for access routers



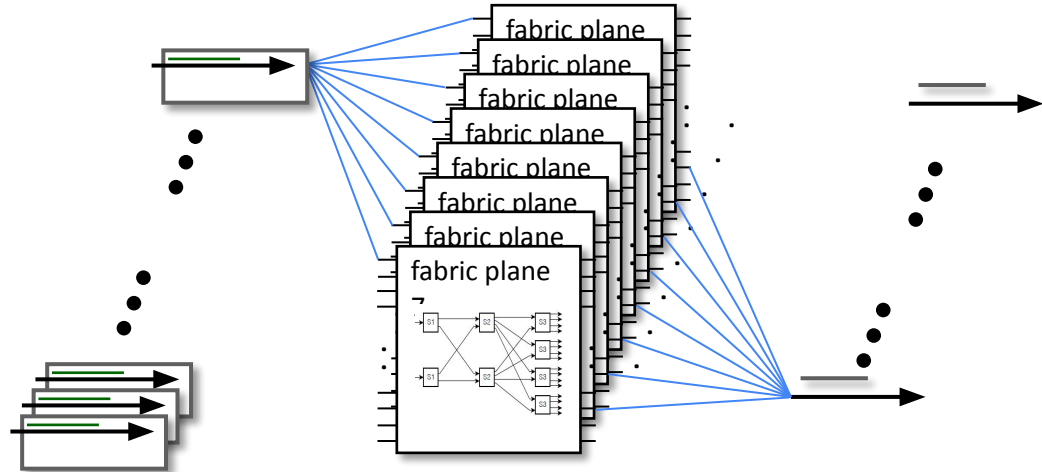
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Switching via interconnection network

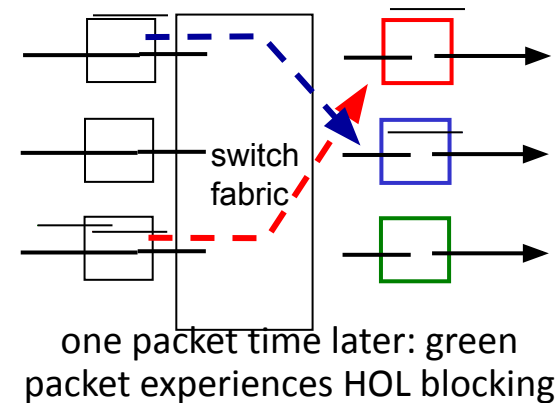
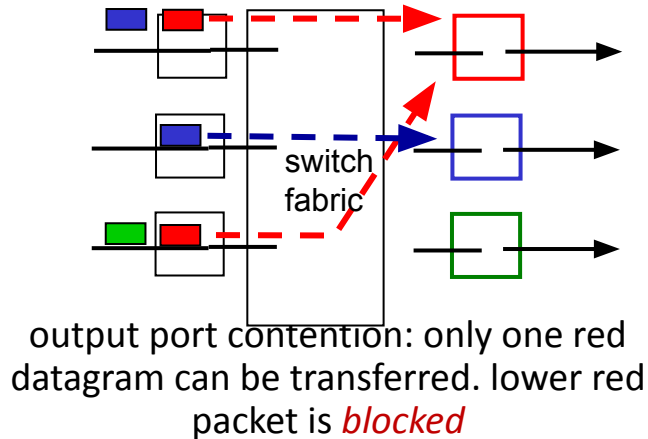
- Crossbar, Clos networks, other interconnection nets initially developed to connect processors in multiprocessor
- **multistage switch**: $n \times n$ switch from multiple stages of smaller switches
- **exploiting parallelism**:
 - fragment datagram into fixed length cells on entry
 - switch cells through the fabric, reassemble datagram at exit



- scaling, using multiple switching “planes” in parallel:
 - speedup, scaleup via parallelism
- Cisco CRS router:
 - basic unit: 8 switching planes
 - each plane: 3-stage interconnection network
 - up to 100’s Tbps switching capacity

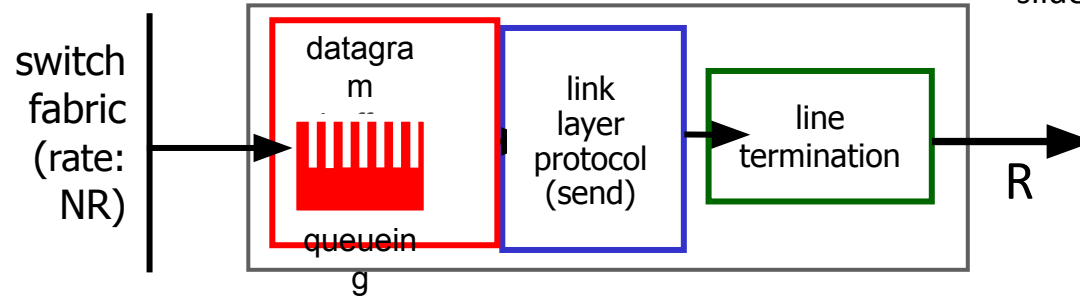


- If switch fabric slower than input ports combined -> queueing may occur at input queues
 - queueing delay and loss due to input buffer overflow!
- **Head-of-the-Line (HOL) blocking:** queued datagram at front of queue prevents others in queue from moving forward





This is a really important
slide



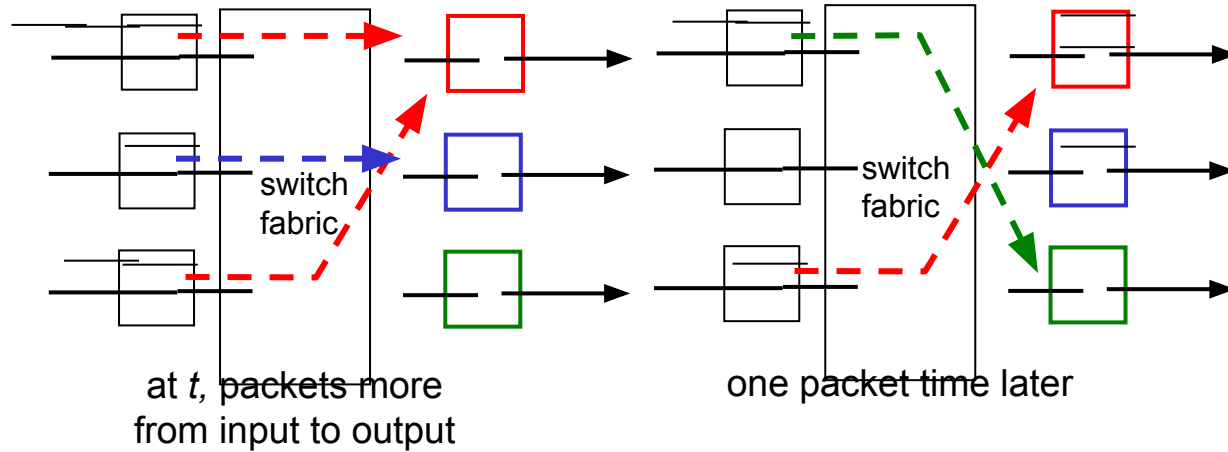
- **Buffering** required when datagrams arrive from fabric faster than link transmission rate. **Drop policy:** which datagrams to drop if no free buffers?
- **Scheduling discipline** chooses among queued datagrams for transmission



Datagrams can be lost due to congestion, lack of buffers



Priority scheduling – who gets best performance, network neutrality

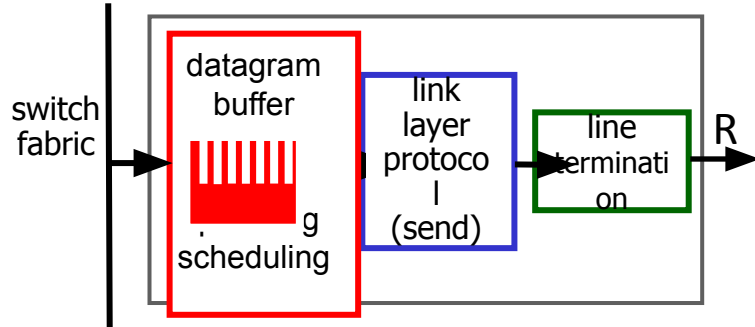


- buffering when arrival rate via switch exceeds output line speed
- *queueing (delay) and loss due to output port buffer overflow!*

- RFC 3439 rule of thumb: average buffering equal to “typical” RTT (say 250 msec) times link capacity C
 - e.g., $C = 10$ Gbps link: 2.5 Gbit buffer
- more recent recommendation: with N flows, buffering equal to

$$\frac{RTT \cdot C}{\sqrt{N}}$$

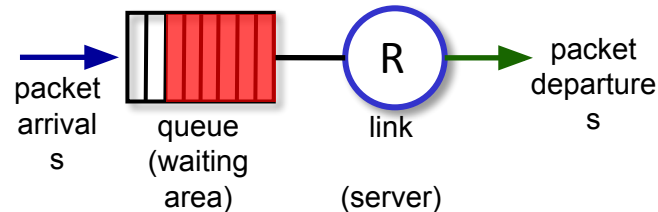
- but *too* much buffering can increase delays (particularly in home routers)
 - long RTTs: poor performance for realtime apps, sluggish TCP response
 - recall delay-based congestion control: “keep bottleneck link just full enough (busy) but no fuller”



buffer management:

- **drop:** which packet to add, drop when buffers are full
 - **tail drop:** drop arriving packet
 - **priority:** drop/remove on priority basis

Abstraction: queue



- **marking:** which packets to mark to signal congestion (ECN, RED)



THANK YOU

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